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NAAC GRADE A++

Networking Services

Study Guide

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Networking Basics

IP Address

Unique numerical label assigned to each device on a network. IPv4: 32-bit (e.g., 192.168.1.1). IPv6: 128-bit.

CIDR Notation

Classless Inter-Domain Routing. Format: IP/prefix (e.g., 10.0.0.0/16). The prefix defines how many bits belong to the network.

Subnet

A logical subdivision of an IP network. Allows network segmentation for security and traffic management.

Gateway

Entry/exit point between networks. In AWS, Internet Gateway (IGW) connects VPC to the public internet.

DNS

Domain Name System — translates human-readable domain names (google.com) to IP addresses (e.g., 172.217.1.1).

NAT

Network Address Translation — allows private subnet instances to access the internet without exposing their private IPs.

OSI Model (Quick Reference)

7	Application	ALB, API Gateway	HTTP, HTTPS, DNS
4	Transport	NLB, Security Groups	TCP, UDP, Ports
3	Network	Route Tables, NACL	IP, ICMP, Routing
2	Data Link	ENI	MAC Address, Ethernet

Key CIDR Quick Facts:

- /16 = 65,536 IP addresses | /24 = 256 IPs | /28 = 16 IPs (smallest AWS subnet)
- AWS VPC supports /16 (largest) to /28 (smallest) CIDR blocks
- AWS reserves 5 IP addresses in every subnet (first 4 + last 1)

```
# CIDR Examples

10.0.0.0/16 → 65,531 usable IPs (65,536 - 5 reserved)

10.0.1.0/24 → 251 usable IPs (256 - 5 reserved)

10.0.1.0/28 → 11 usable IPs (16 - 5 reserved) — SMALLEST

10.0.0.0/8 → Private (Class A)
```

```
172.16.0.0/12 → Private (Class B)

192.168.0.0/16 → Private (Class C)
```

Virtual Private Cloud (VPC)

What is a VPC?

Amazon VPC lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network you define. You have complete control over your virtual networking environment — including IP address ranges, subnet creation, route table configuration, and network gateway settings.

`Default VPC`

Auto-created when you first use EC2. Has one public subnet per AZ, an internet gateway, and a main route table. CIDR: 172.31.0.0/16.

`Custom VPC`

Created manually. You define the CIDR block, subnets, routing, and gateways. Best practice for production workloads.

Key VPC Components

- **Internet Gateway (IGW)** — Connects VPC to internet (1 per VPC)
- **NAT Gateway** — Private subnet internet access (no inbound)
- **Elastic IP (EIP)** — Static public IPv4 address
- **Elastic Network Interface (ENI)** — Virtual NIC attached to EC2
- **VPC Peering** — Connect two VPCs (no overlapping CIDRs)
- **VPC Endpoints** — Private access to AWS services
- **Transit Gateway** — Hub-and-spoke VPC connectivity
- **VPN Gateway (VGW)** — Site-to-site VPN connection

📌 **VPC Limits (Default):** 5 VPCs per region · 200 subnets per VPC · 5 IGWs per region · VPC CIDR range: /16 to /28

✅ **Exam Tip:** A VPC spans all Availability Zones in a region. Subnets exist in only ONE Availability Zone. An IGW is automatically highly available.

Subnets

🌐 Public Subnet

- Has a route to an Internet Gateway
- Resources can have public IPs
- Hosts: Web servers, Load Balancers, Bastion Hosts, NAT Gateways
- Direct internet access possible

🔒 Private Subnet

- No direct route to Internet Gateway
- Resources use NAT Gateway for outbound internet
- Hosts: Databases, App servers, Backend services
- More secure — no inbound from internet

⚠ **Remember:** Subnets are AZ-specific resources. Each subnet exists in exactly ONE Availability Zone. To achieve high availability, create subnets in multiple AZs.

Subnet IP Reservation by AWS (5 addresses per subnet)

1st IP	Network address	10.0.1.0
2nd IP	VPC router	10.0.1.1
3rd IP	DNS server	10.0.1.2
4th IP	Future use (reserved)	10.0.1.3
Last IP	Broadcast address	10.0.1.255

Route Tables

What are Route Tables?

A route table contains a set of rules (routes) that determine where network traffic is directed. Every VPC has a main route table by default. Each subnet must be associated with a route table.

Public Route Table

10.0.0.0/16	local
0.0.0.0/0	igw-xxxxxxxx

The 0.0.0.0/0 → IGW route enables internet access

Private Route Table

10.0.0.0/16	local

0.0.0.0/0	nat-xxxxxxxx

0.0.0.0/0 → NAT Gateway for outbound-only internet
✅ **Key Rule:** The most specific route wins (longest prefix match). A /32 route overrides /24 which overrides /0. The "local" route always exists and cannot be deleted — it allows internal VPC communication.

Security Groups & VPC Security

Security Groups (Instance-Level Firewall)

- Acts as a virtual firewall for EC2 instances
- Operates at the **instance level** (not subnet level)
- Supports only **ALLOW** rules — no deny rules
- **Stateful** — if you allow inbound, outbound response is automatically allowed
- New security groups: all inbound denied, all outbound allowed by default
- Multiple security groups can be assigned to one instance

Network Access Control Lists (NACLs)

- Optional layer at the **subnet level**
- Supports both **ALLOW and DENY** rules
- **Stateless** — return traffic must be explicitly allowed
- Rules evaluated in number order (lowest to highest); first match wins
- Default NACL: allows all inbound and outbound traffic
- Custom NACL: denies all inbound and outbound by default

Operates at	Instance level	Subnet level
Rules	Allow only	Allow & Deny
Statefulness	Stateful	Stateless
Rule evaluation	All rules evaluated	Rules in number order
Default (new)	Deny inbound, allow outbound	Depends (default allows all)
Association	EC2 instances	Subnets

Common Security Group Port Rules

22	TCP	SSH	Inbound
80	TCP	HTTP	Inbound
443	TCP	HTTPS	Inbound
3306	TCP	MySQL/Aurora	Inbound (from app SG)
5432	TCP	PostgreSQL	Inbound (from app SG)
3389	TCP	RDP (Windows)	Inbound
1024–65535	TCP	Ephemeral ports	NACL outbound (stateless)

VPC Flow Logs

Capture information about IP traffic going to/from network interfaces in your VPC. Can be published to CloudWatch Logs or S3. Used for: security analysis, compliance, troubleshooting connectivity issues. Available at VPC level, subnet level, or ENI level.

⚡ **VPC Peering Security:** Security groups in a peered VPC can be referenced by name. NACLs operate independently — each subnet's NACL applies to that subnet's traffic only. VPC peering is NOT transitive (A↔B, B↔C does NOT mean A↔C).

DNS: AWS Route 53

What is Route 53?

Amazon Route 53 is a highly available, scalable, and fully managed Domain Name System (DNS) web service. It translates domain names into IP addresses. Named after TCP/UDP port 53 (DNS port). It's also a domain registrar.

Route 53 Routing Policies

Simple	Single resource	Returns single record value; random if multiple values
Weighted	A/B testing, gradual migration	Splits traffic by weight (e.g., 70% / 30%)
Latency-Based	Global apps, low latency	Routes to region with lowest latency for user
Failover	Active-passive DR	Routes to secondary if health check fails on primary

Geolocation	Geo-restriction, localization	Routes based on user's geographic location
Geoproximity	Bias-based routing	Routes based on resource/user locations with bias factor
Multivalue Answer	Random load distribution	Returns up to 8 healthy records randomly
IP-Based	ISP-specific routing	Routes based on client IP address/CIDR

Hosted Zone

Container for DNS records for a domain. Two types: Public (internet-facing) and Private (within VPC only).

Health Checks

Route 53 monitors endpoints. If unhealthy, traffic rerouted. Supports HTTP, HTTPS, TCP. Used with failover routing.

Alias Record

AWS-specific. Points to AWS resources (ELB, CloudFront, S3). Free of charge. Works at zone apex (root domain).

TTL (Time to Live)

Time DNS resolvers cache a record. Lower TTL = faster propagation but more DNS queries. Higher TTL = fewer queries.

DNS Record Types

`A`	Maps domain to IPv4 address	example.com → 1.2.3.4
`AAAA`	Maps domain to IPv6 address	example.com → 2001::1
`CNAME`	Maps domain to another domain	www → example.com
`MX`	Mail exchange records	Email routing
`NS`	Name servers for hosted zone	Delegation
`TXT`	Text information (verification)	SPF, DKIM records
`PTR`	Reverse DNS lookup	IP → domain name
`SOA`	Start of Authority	Zone metadata

✅ **Exam Tip:** CNAME cannot be used for the zone apex (root domain like "example.com") — use an Alias record instead. Alias records can point to: ELB, CloudFront, S3 static website, Elastic Beanstalk, API Gateway, VPC endpoints.

Amazon CloudFront (CDN)

What is CloudFront?

Amazon CloudFront is a Content Delivery Network (CDN) service that securely delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds. It uses a global network of **Edge Locations** to cache content closer to users.

`Edge Locations`

Data centers where CloudFront caches content. More numerous than AZs/Regions (~400+ globally). Content served from nearest edge location.

`Origin`

The source of the content: S3 bucket, EC2 instance, ELB, or any HTTP server. CloudFront fetches content from origin on cache miss.

`Distribution`

A CloudFront configuration consisting of origins, behaviors, and settings. Two types: Web distribution and RTMP (deprecated).

`Cache Behavior`

Rules that define how CloudFront handles requests for specific URL paths — TTL, caching, HTTP methods allowed.

CloudFront Key Features

- **HTTPS:** SSL/TLS termination at edge
- **WAF Integration:** Filter malicious requests
- **Geo-Restriction:** Block/allow by country
- **Signed URLs/Cookies:** Restrict access to private content
- **Origin Shield:** Additional caching layer
- **Lambda@Edge:** Run code at edge locations
- **Field-Level Encryption:** Protect sensitive data
- **DDoS Protection:** Integrated with AWS Shield

Cache static content (images, CSS, JS)	CloudFront + S3 origin
Reduce latency for global users	CloudFront edge locations
Protect against DDoS	CloudFront + AWS Shield + WAF
Stream video	CloudFront + S3/MediaPackage
Restrict content to paid users	CloudFront Signed URLs/Cookies

Serve both dynamic & static content	CloudFront with multiple origins

⚡ **CloudFront vs Route 53 vs Global Accelerator:**

- **CloudFront** — CDN, caches content at edge locations, HTTP/HTTPS
- **Route 53** — DNS service, routes requests to endpoints worldwide
- **Global Accelerator** — Routes TCP/UDP traffic using AWS backbone network (no caching)

Build Your VPC & Launch a Web Server (Lab)

Architecture Overview

Build a 2-tier VPC: public subnet (web server) + private subnet (database). Internet access via IGW for public, NAT Gateway for private.

Create VPC

Go to VPC Console → Create VPC → Name: MyVPC → IPv4 CIDR: 10.0.0.0/16 → Tenancy: Default → Create

Create Subnets

Public Subnet: CIDR 10.0.1.0/24, AZ: us-east-1a. **Private Subnet:** CIDR 10.0.2.0/24, AZ: us-east-1b. Enable auto-assign public IP on public subnet.

Create & Attach Internet Gateway

VPC → Internet Gateways → Create IGW → Name: MyIGW → Attach to MyVPC. One IGW per VPC only.

Configure Route Tables

Public RT: Add route 0.0.0.0/0 → IGW. Associate with public subnet.
Private RT: Add route 0.0.0.0/0 → NAT Gateway. Associate with private subnet.

Create NAT Gateway

VPC → NAT Gateways → Create → Place in PUBLIC subnet → Allocate Elastic IP → Create. Takes ~5 min to become available.

Create Security Groups

Web-SG: Inbound: HTTP (80), HTTPS (443), SSH (22) from your IP.
DB-SG: Inbound: MySQL (3306) from Web-SG only (reference SG name).

Launch EC2 Web Server

EC2 → Launch Instance → AMI: Amazon Linux 2 → Place in public subnet → Assign Web-SG → Add User Data script to install Apache (httpd).

User Data Script (Apache)

```
#!/bin/bash

yum update -y

yum install -y httpd
```

```
systemctl start httpd

systemctl enable httpd

echo "<h1>Hello from AWS VPC!</h1>" > /var/www/html/index.html
```

Test the Web Server

Copy the EC2 Public IPv4 address → Open browser → http://<public-ip> → You should see "Hello from AWS VPC!" page.

(Optional) Add Route 53 + CloudFront

Register domain in Route 53 → Create CloudFront distribution with EC2/ELB as origin → Create Route 53 A Alias record pointing to CloudFront distribution domain.

✅ **Remember:** NAT Gateway must be placed in a PUBLIC subnet. It requires an Elastic IP. Private subnets route outbound internet traffic through NAT Gateway — inbound connections from internet are blocked.

UNIT I — NETWORKING & VPC BASICS

What is Amazon VPC and why is it used? Explain its key components.

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ANSWER Amazon Virtual Private Cloud (VPC) is a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network you define.

Why used: Provides complete control over network configuration — IP ranges, subnets, route tables, gateways, and security settings. Mimics a traditional on-premises data center network.

Key Components:

- **Subnets** — Divide VPC's IP range into smaller segments (public/private)
- **Internet Gateway (IGW)** — Enables internet connectivity for public subnets
- **Route Tables** — Control traffic routing within VPC
- **Security Groups** — Instance-level stateful firewall
- **NACLs** — Subnet-level stateless firewall
- **NAT Gateway** — Outbound internet for private subnets
- **VPC Endpoints** — Private access to AWS services without internet

What is CIDR notation? If a VPC has CIDR block 10.0.1.0/24, how many IP addresses are available for use?

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ANSWER **CIDR (Classless Inter-Domain Routing)** notation represents an IP address and its associated network mask. Format: IP_address/prefix_length.

For **10.0.1.0/24**:
Total IPs = 2^(32-24) = 2^8 = 256

AWS reserves 5 IPs in every subnet (1st four + last one)
Usable IPs = 256 - 5 = 251

The /24 prefix means the first 24 bits define the network (10.0.1) and the last 8 bits are for hosts (0-255).

Differentiate between Public Subnet and Private Subnet with examples of what to host in each.

▼

ANSWER

Internet Gateway Route	Yes (0.0.0.0/0 → IGW)	No direct IGW route
Public IP	Instances can have public IPs	No public IPs
Internet Access	Direct (inbound + outbound)	Outbound only via NAT
Security Level	Lower (internet-facing)	Higher (isolated)
Hosts	Web servers, ALB, Bastion, NAT GW	Databases, App servers, Cache

What is the smallest subnet CIDR block allowed in an AWS VPC? How many usable IPs does it provide?

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ANSWER The smallest subnet allowed in AWS is **/28**.

Total IPs in /28 = 2^(32-28) = 2^4 = **16 IP addresses**
AWS reserves 5 IPs (first 4 + last 1)
Usable IPs = 16 - 5 = 11

The largest allowed subnet in a VPC is **/16** (65,536 total IPs, 65,531 usable).

Explain VPC Peering. What are its limitations?

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ANSWER **VPC Peering** is a networking connection between two VPCs that enables routing of traffic between them using private IPv4/IPv6 addresses.

Key Limitations:

- **No transitive peering** — if A↔B and B↔C, A cannot communicate with C through B
- **No overlapping CIDRs** — peered VPCs cannot have the same IP ranges
- **No edge-to-edge routing** — VPN/Direct Connect of one VPC is not accessible from the other
- Works across accounts and regions (inter-region peering supported)
- You must update route tables in both VPCs
- Security groups of peered VPC can be referenced by name

UNIT II — SECURITY GROUPS & NACLS

Compare Security Groups and Network ACLs (NACLs). Which is stateful and which is stateless? Why does it matter?

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ANSWER **Security Groups = Stateful:** When you allow inbound traffic, the response is automatically allowed outbound regardless of outbound rules. AWS tracks the connection state.

NACLs = Stateless: Each request is evaluated independently. If you allow inbound HTTP (port 80), you must also explicitly allow outbound ephemeral ports (1024–65535) for the response to reach the client. This is because NACLs don't track connection state.

Why it matters: With NACLs, forgetting to configure ephemeral ports for return traffic will break connectivity even if the inbound rule is correct. Security Groups are simpler because return traffic is automatic.

What happens by default when you create a new Security Group in AWS?

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ANSWER When you create a new (custom) Security Group:

- **Inbound:** All traffic is **DENIED** by default (no inbound rules)
- **Outbound:** All traffic is **ALLOWED** by default (rule: 0.0.0.0/0 all traffic allowed)

This means instances with a new security group can initiate outbound connections but accept no inbound connections until you explicitly add inbound rules.

What are VPC Flow Logs? What information do they capture and where can they be stored?

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ANSWER **VPC Flow Logs** capture metadata about IP traffic flowing through your VPC network interfaces.

Information captured: Source/destination IP, source/destination port, protocol, packet count, byte count, start/end time, action (ACCEPT/REJECT), log status.

Storage destinations:

- Amazon CloudWatch Logs (for querying with Logs Insights)
- Amazon S3 (for cost-effective long-term storage and Athena queries)
- Amazon Kinesis Data Firehose (for real-time streaming)

Available at: VPC level, Subnet level, or individual ENI level. Flow logs don't capture: DNS queries, DHCP traffic, instance metadata (169.254.169.254) traffic.

A company wants to allow all instances in a private subnet to access the internet for software updates, but no inbound traffic from the internet. What AWS component is needed?

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ANSWER A **NAT Gateway (Network Address Translation Gateway)** is needed.

How it works:

- Place NAT Gateway in the **PUBLIC subnet** (it needs internet access)
- Assign an **Elastic IP** to the NAT Gateway
- Update the **private subnet's route table**: 0.0.0.0/0 → NAT Gateway
- Private instances initiate outbound connections; responses flow back through NAT
- Internet cannot initiate connections TO private instances

Note: NAT Gateway is managed and highly available (within an AZ). For cross-AZ HA, deploy one NAT GW per AZ.

UNIT III — ROUTE 53 (DNS)

Explain the different routing policies in AWS Route 53 with use cases for each.

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ANSWER
- **Simple:** Single resource; no health checks. Use for basic routing. If multiple values, returns all (client picks randomly).
- **Weighted:** Distribute traffic by percentage. Use for A/B testing, blue/green deployments (e.g., 90% to v1, 10% to v2).
- **Latency-Based:** Routes to lowest-latency region for the user. Best for global applications.
- **Failover:** Primary + secondary resources. If health check fails on primary, traffic shifts to secondary. Use for active-passive disaster recovery.
- **Geolocation:** Routes based on user's country/continent. Use for legal compliance, localized content.
- **Geoproximity:** Routes based on geographic distance with adjustable bias. Requires Route 53 Traffic Flow.
- **Multivalue Answer:** Returns up to 8 healthy records randomly. Similar to simple but with health checks.
- **IP-Based:** Route based on client IP CIDR. Use for ISP-specific routing.

What is the difference between a CNAME record and an Alias record in Route 53?
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ANSWER

Points to	Any domain name	AWS resources only
Zone apex	Cannot be used (e.g., example.com)	Can be used at root domain
Cost	Charged per query	Free for AWS resources
TTL	You set the TTL	Managed by AWS automatically
Targets	Any hostname	ELB, CloudFront, S3, API GW, etc.
Health check	Not supported natively	Inherits target health

Key rule: At the zone apex (root domain like "example.com"), you MUST use an Alias record, not CNAME.

A company wants to route 70% of traffic to a new application version and 30% to the old version. Which Route 53 routing policy should be used?
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ANSWER Use **Weighted Routing Policy**.

Configuration:
- Create Record Set 1: Points to new app → Weight: 70
- Create Record Set 2: Points to old app → Weight: 30
- Route 53 sends 70/(70+30) = 70% traffic to new app

Use cases: A/B testing, canary deployments, gradual migration, blue/green deployments. If a weighted record's value is 0, no traffic is sent to it. If all records have weight 0, traffic is distributed equally.

What is a Hosted Zone in Route 53? Differentiate between Public and Private Hosted Zones.

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ANSWER A **Hosted Zone** is a container for DNS records belonging to a domain. It defines how Route 53 responds to DNS queries for a domain.

Public Hosted Zone:

- Contains records for publicly accessible domain names
- Responds to DNS queries from the public internet
- Example: example.com resolving to a public IP

Private Hosted Zone:

- Contains records for domain names used within VPCs
- Only responds to DNS queries from within associated VPCs
- Must associate with one or more VPCs
- Supports split-view DNS (same domain name, different IP inside/outside VPC)

UNIT IV — CLOUDFRONT & CDN

What is Amazon CloudFront? How does it improve performance and what are Edge Locations?

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ANSWER **Amazon CloudFront** is AWS's Content Delivery Network (CDN) that delivers data, videos, applications, and APIs globally with low latency.

How it improves performance:

- Caches content at Edge Locations (400+ globally) near users
- User requests are served from the nearest edge location instead of the origin server
- Reduces latency — especially for static content (images, CSS, JS, videos)
- Uses AWS backbone network for communication between edge and origin

Edge Locations: Data centers distributed worldwide where CloudFront caches copies of content. More numerous than AWS Regions and AZs. When a user requests content, Route 53 routes them to the nearest edge location. If cached (cache hit) → served immediately. If not (cache miss) → fetched from origin server.

How does CloudFront handle HTTPS and what is the benefit of integrating it with AWS WAF?

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ANSWER **CloudFront + HTTPS:**

- CloudFront terminates HTTPS at the edge location (SSL/TLS offloading)
- You can use AWS Certificate Manager (ACM) for free SSL certificates
- Between edge and origin: can use HTTP or HTTPS (Origin Protocol Policy setting)
- Supports SNI (Server Name Indication) for virtual hosting

CloudFront + AWS WAF (Web Application Firewall):

- WAF rules applied at edge locations — block malicious traffic before it reaches origin
- Protection against: SQL injection, XSS, DDoS, bot attacks, rate limiting
- Geoblocking: restrict content by country at edge level
- Reduces load on origin server — bad traffic never reaches it

What are CloudFront Signed URLs and Signed Cookies? When would you use each?

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ANSWER Both mechanisms restrict access to private CloudFront content:

Signed URLs:
- One URL grants access to one specific file
- Contains expiry time, IP restrictions, and cryptographic signature
- Use when: distributing individual files (e.g., paid PDF download, single video)
- Use when: user's app doesn't support cookies

Signed Cookies:
- One cookie grants access to multiple files/URLs
- User doesn't see complex query strings in URLs
- Use when: providing access to all files in a restricted area (e.g., subscriber video library)
- Use when: you don't want to change URLs

UNIT V — SCENARIO & ARCHITECTURE QUESTIONS

Describe the steps to build a VPC with a public subnet (web server) and private subnet (database) in AWS.

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ANSWER **Step-by-Step:**

1. Create VPC with CIDR 10.0.0.0/16
2. Create Public Subnet (10.0.1.0/24 in AZ-a) — enable auto-assign public IP
3. Create Private Subnet (10.0.2.0/24 in AZ-b)
4. Create Internet Gateway → attach to VPC
5. Create Public Route Table → add 0.0.0.0/0 → IGW → associate with public subnet
6. Create NAT Gateway in public subnet with Elastic IP
7. Create Private Route Table → add 0.0.0.0/0 → NAT GW → associate with private subnet
8. Create Web-SG: allow HTTP(80), HTTPS(443), SSH(22) inbound
9. Create DB-SG: allow MySQL(3306) inbound from Web-SG only
10. Launch EC2 (web server) in public subnet with Web-SG; install Apache via User Data
11. Launch RDS/EC2 (database) in private subnet with DB-SG
12. Test: Access web server's public IP in browser

A global e-commerce company wants to reduce page load time for international users. What combination of AWS services would you recommend?

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ANSWER **Recommended Architecture:**

- **Route 53 with Latency-Based Routing** — directs users to the nearest region's infrastructure
- **CloudFront CDN** — caches static content (images, CSS, JS) at 400+ edge locations globally
- **S3 + CloudFront** — serve static assets (product images, catalog pages)
- **ALB in each region** — distribute traffic to EC2 instances or containers
- **VPC per region** — isolated network environments
- **AWS Global Accelerator** — if low-latency routing is needed for non-HTTP (TCP/UDP) traffic or dynamic content

Key benefit: CloudFront reduces latency by ~50-70% for static content. Route 53 latency routing ensures users connect to the fastest region.

What is a Bastion Host (Jump Server) in AWS VPC? How is it configured?

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ANSWER A **Bastion Host** (also called Jump Server or Jump Box) is a special-purpose EC2 instance used to provide secure SSH/RDP access to instances in private subnets.

Configuration:

- Placed in a **public subnet** with a public IP or Elastic IP
- Security Group: Allow inbound SSH (port 22) only from your specific IP or corporate IP range
- Private instances: Security Group allows SSH inbound from the Bastion Host's Security Group
- You SSH into Bastion first, then SSH from Bastion to private instances

Best Practice Alternative: Use AWS Systems Manager Session Manager (SSM) — provides browser-based shell access without needing a Bastion host or open SSH ports.

What is an AWS VPC Endpoint? Differentiate between Interface Endpoint and Gateway Endpoint.

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ANSWER A **VPC Endpoint** allows private connectivity between your VPC and supported AWS services without requiring internet, VPN, NAT, or Direct Connect — traffic stays within the AWS network.

Gateway Endpoint:

- Supports: Amazon S3 and DynamoDB only
- Added as a target in route table
- Free of charge
- Works within same region

Interface Endpoint (powered by AWS PrivateLink):

- Supports: 100+ AWS services (EC2, SSM, Secrets Manager, KMS, etc.)
- Creates an Elastic Network Interface (ENI) in your subnet with a private IP
- Charged per hour + per GB data processed
- Uses DNS to route traffic to the endpoint

Explain the difference between Internet Gateway and NAT Gateway. When would you use each?

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ANSWER

Purpose	Connect VPC to internet	Outbound internet for private subnets
Direction	Inbound + Outbound	Outbound only (no inbound)
Placed in	Attached to VPC	Public subnet
Used with	Public subnets	Private subnets
Elastic IP	Not required	Required
High Availability	Inherently HA (AWS managed)	AZ-level HA; deploy per AZ for full HA
Cost	Free (data transfer charges)	Hourly charge + data processing charge

What is Route 53 Health Checking? How does it integrate with routing policies?

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ANSWER **Route 53 Health Checks** monitor the availability and performance of your endpoints (web servers, other resources).

Types of health checks:
- **Endpoint monitoring:** Checks HTTP/HTTPS/TCP endpoint health
- **Calculated health checks:** Combines results of multiple health checks
- **CloudWatch alarms:** Based on CloudWatch metric alarm state

Integration with routing:
- **Failover routing:** Health check on primary; if fails → traffic to secondary
- **Weighted/Latency/Geolocation:** Only healthy endpoints receive traffic
- **Multivalue Answer:** Returns only healthy records (up to 8)

Health checkers are globally distributed — a resource is healthy only when a percentage (default 18%) of global checkers can reach it.

Write the steps to configure HTTPS for a web application using CloudFront + Route 53 + ACM.

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ANSWER

1. Register your domain in Route 53 (or transfer existing domain)
2. Go to **AWS Certificate Manager (ACM)** → Request a public certificate → Enter domain name (e.g., example.com, *.example.com)
3. Validate domain ownership via DNS validation — ACM creates a CNAME record in Route 53 automatically
4. Wait for certificate status to become **Issued**
5. Create **CloudFront distribution** → Set origin (EC2/ELB/S3)

6. In CloudFront: Viewer Protocol Policy → "Redirect HTTP to HTTPS"

7. Alternate Domain Names (CNAMEs): enter your domain (example.com)

8. SSL Certificate: select your ACM certificate (must be in us-east-1)

9. Create distribution → copy the CloudFront domain name (xxxx.cloudfront.net)

10. In **Route 53** → Hosted Zone → Create A record → Alias → Select CloudFront distribution

11. Test: https://example.com — should show padlock with valid SSL

Note: ACM certificates for CloudFront MUST be created in the us-east-1 (N. Virginia) region.

What is the Default VPC in AWS? Can you delete it? What are its default configurations?

▼

ANSWER The **Default VPC** is automatically created in every AWS region when you first use the EC2 service.

Default configurations:

- CIDR block: 172.31.0.0/16
- One public subnet per Availability Zone (with /20 CIDR each)
- An Internet Gateway attached
- A main route table with 0.0.0.0/0 → IGW (all subnets are public by default)
- Default Security Group, NACL, and DHCP options set
- Auto-assign public IPv4 enabled on all subnets

Can you delete it? Yes, you can delete the default VPC. AWS also allows you to recreate it via the console or AWS CLI. Best practice: create a custom VPC for production and avoid using the default VPC.

A user cannot reach their EC2 web server via the public IP. List 5 things you would check to debug this issue.

▼

ANSWER **Troubleshooting checklist:**

1. **Security Group:** Does the inbound rule allow HTTP (port 80) or HTTPS (443) from 0.0.0.0/0?

2. **Route Table:** Does the subnet's route table have a route 0.0.0.0/0 → Internet Gateway?

3. **Internet Gateway:** Is an IGW created and attached to the VPC?

4. **Public IP:** Does the EC2 instance have a public IP address or Elastic IP assigned?

5. **NACL:** Does the subnet's Network ACL allow inbound HTTP/HTTPS and outbound ephemeral ports (1024–65535)?

6. **EC2 status:** Is the instance in "running" state? Is the web server (Apache/Nginx) actually running?

7. **Subnet:** Is the EC2 instance in a public subnet (not a private subnet)?

8. **VPC Flow Logs:** Check if traffic is being REJECTED at any layer

`SHORT ANSWER & MCQ-STYLE QUESTIONS`

Quick MCQ Points to Remember

- VPC spans: `All AZs in a region` | Subnets span: `ONE AZ only`
- Smallest subnet in VPC: `/28` | Largest: `/16`
- AWS reserves: `5 IPs per subnet` (first 4 + last 1)
- Security Groups are: `Stateful, Allow-only` | NACLs are: `Stateless, Allow+Deny`
- NAT Gateway placed in: `PUBLIC subnet`
- IGW: `1 per VPC` | Can be shared? `No`
- VPC Peering is: `NOT transitive`
- CloudFront caches at: `Edge Locations` (not Regions)
- Route 53 is: `Global service` (not regional)
- ACM certificate for CloudFront must be in: `us-east-1`
- Route 53 record for root domain: `Alias (not CNAME)`
- Gateway Endpoint supports: `S3 and DynamoDB only`
- VPC Flow Logs do NOT capture: `DNS traffic, DHCP, metadata (169.254.x.x)`
- Default VPC CIDR: `172.31.0.0/16`

AWS Networking Study Notes — Prepared for University Exam Preparation · Covers VPC, Subnets, Route Tables, Security Groups, Route 53, CloudFront